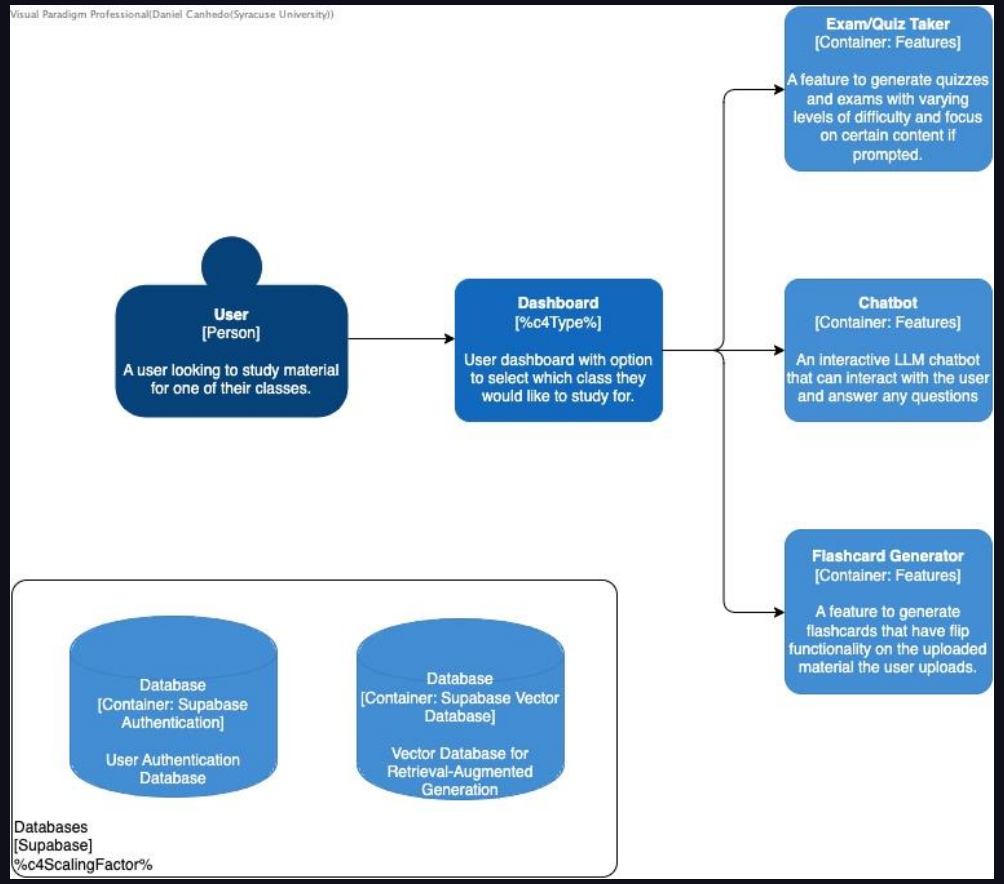
CIS MIDTERM PRESENTATION

StudyBuddy — AI-Powered Study Assistant

An AI tutor that lets students upload course materials and study through chat, flashcards, and quizzes — all grounded in their own class content.

Techstack: HTML/CSS, JavaScript, Python

Team: Daniella Lat • Daniel Canhedo • Allison Simpson • Azka Naseem



SPRINT METRICS

Where we stand by the numbers

24	3	6	0.75
STORIES COMPLETED	STORIES ROLLED OVER	AVG COMPLETED / SPRINT	AVG ROLLED / SPRINT

Story Points per Member

Member	Hours
Daniella Lat	~44h 35m
Daniel Canhedo	~40h 30m
Allison Simpson	~34h 20 m
Azka Naseem	~44h 51m

Why Stories Rolled Over

→ pgvector took longer to set up than expected for RAG implementation

→ UI/UX design changed a few times so we had to reconvene on one design

→ Had to spend extra time on documentation. Didn’t do a good job in early sprints at documentation.

What went well & what we'd improve

What Went Well

→ Project directory structure was followed from the start. Helps to keep our project organized

→ FastAPI backend stood up quickly, clean router design

→ Initial LLM chain works well with given prompts

→ Frontend quiz UI came together fast with solid UX polish

→ Team communication has been well within Jira and during SCRUM meetings, helping to keep things moving

Could Have Gone Better

→ Sprint organization could have been better during sprint 1 and 2. Took some time to get used to a flow but now things are moving smoothly.

→ Git workflow caused merge conflicts early.

→ Didn't start documentation early enough — had to backfill and spent time writing documentation

→ Work on implementing test cases for certain app-critical functions

System architecture & what we're delivering

Frontend

HTML / CSS / JS

Quiz UI, Chat, Dashboard

Login & Signup

→

FastAPI Backend

Routes: /chat, /quiz, /study

Agent orchestrator

Prompt templates

→

AI + Data Layer

OpenAI Agents SDK

RAG retrieval (embeddings)

PostgreSQL + PyPDF

Our release demo will show the end-to-end flow: a student uploads a PDF, asks a question via chat, generates flashcards, and takes a quiz — all with context of their uploaded material.

Are we on track? (main functionalities)

Original Plan

- → Upload course materials (PDF, DOCX)
- → AI chat grounded in student content
- → Flashcard generation from materials
- → Quiz generation + scoring + weak topic analysis
- → User auth + dashboard

Current Status

WIP	PDF/DOCX upload + chunking
WIP	Chat mode with RAG
WIP	Auth + dashboard wiring
DONE	Quiz generation + results UI
DONE	Flashcard generation

DOCUMENTATION

How we track issues, designs & artifacts

Issues & Challenges

Tracked in Jira with labels for bugs, blockers, and tech debt. Sprint board + backlog keep everything visible.

Design Artifacts

Architecture diagrams, prompt templates, and UI wireframes stored in our Github repo and linked from Jira stories.

Jira Sprint Story

Each team member has a story in the current sprint for this midterm presentation, with subtasks per slide.

Key Documented Artifacts:

System Architecture Diagram • API Route Map • Prompt Templates • UI • Sprint Retro Notes • README + Setup Guide

StudyBuddy Pics

Create your account.

Sign up to save progress, quizzes, and study plans.

Email

you@syr.edu

Password

Create a password

Confirm password

Repeat password

Sign Up | Back

StudyBuddy
Quiz

Azka

Automata practice quiz

Answer a few multiple-choice questions. When you're done, you'll see your score, a breakdown by question, and which topics to review.

Quiz summary

Automata | Medium | 3 questions

Enable timer

Time limit (minutes) 10

If enabled, a countdown starts when the quiz begins. When time is up, your quiz is automatically submitted.

Start quiz

STUDYBUDDY

Deck Not Found

The deck you're looking for doesn't exist or couldn't be loaded.

← Back to Homepage

Automata quiz

3 questions • Medium

Topic: Mixed

Question 1 / 3 | DFA | No timer

In a DFA, for each state and input symbol, how many transitions are allowed?

D = deterministic

- 0 — Option A
- 1 — Option B
- 2 — Option C
- Any number — Option D

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← Homepage | Ask AI

My Flashcard Decks

Choose a deck to study, or create a new one to get started.

Sort by: Name (A → Z)

7 decks — A → Z

+ Create New Deck

Data Structures Basics
Computer Science 101
Modified Apr 9, 2025
TOTAL CARDS 24 | MASTERED 18

Design Process
Engineering Design
Modified Mar 19, 2025
TOTAL CARDS 12 | MASTERED 8

Integration Techniques
Calculus II
Modified Apr 13, 2025
TOTAL CARDS 30 | MASTERED 22

Kinematics
Physics I
Modified Apr 14, 2025
TOTAL CARDS 25 | MASTERED 20

Limits and Derivatives
Calculus II
Modified Mar 27, 2025
TOTAL CARDS 20 | MASTERED 15

Newton's Laws
Physics I
Modified Apr 7, 2025
TOTAL CARDS 18 | MASTERED 12

Sorting Algorithms
Computer Science 101
Modified Mar 31, 2025
TOTAL CARDS 15 | MASTERED 10

THANK YOU

Questions?

We're happy to demo any part of the system live or walk through specific design decisions.

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